A systematic filter for polynomial continued fractions with factorial reduction: hidden Euler-type cases and fifteen irreducible candidates

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Abstract

We describe a reproducible pipeline that searches polynomial continued fractions (PCFs) with quadratic partial denominators and quartic partial numerators, keeps those with full factorial reduction (FR), and removes every case explained by known mechanisms: the family of Conjecture 1.3 of Ben David et al., deflation, rational or quadratic limits, Möbius transforms of standard constants and of the limits in H. Cohen's databases, simple ${}_3F_2$ values, and Euler-type PCFs detected through hypergeometric solutions of the three-term recurrence. A PCF that passed all numerical identification tests turns out to be of Euler type in disguise: its numerators have the closed form $P_n = -(n+1)!\prod_{m=0}^{n-1}(3m+2)\,(n^2+3n+3)$, which gives its limit as an explicit series and explains its gcd structure. Applying the full filter to 5,274 FR cases leaves fifteen PCFs whose associated order-four Fuchsian operator is irreducible over $\mathbb{Q}(z)$ and whose limits match none of 21,984 tested constants at 1,000 digits. We publish them as well-tested candidates, not as new constants, together with all code and data. Two further members of the same families have limits that are Möbius transforms of $\Gamma(1/3)$-type periods.

1 Conventions

A PCF is $x = a(0) + \dfrac{b(1)}{a(1) + \dfrac{b(2)}{a(2) + \cdots}}$ with $a, b \in \mathbb{Z}[n]$; P_n, Q_n satisfy $u_n = a(n)u_{n-1} + b(n)u_{n-2}$ with $P_{-1} = 1$, $P_0 = a(0)$, $Q_{-1} = 0$, $Q_0 = 1$. The PCF has full FR if $S_N = \frac{1}{N}\log(|Q_N|/\gcd(P_N, Q_N))$ stays bounded; numerically we require $|S_{800} - S_{200}| < 0.15$ (exact integers).

2 Search and filters

1. Brute force over about 2.3×10^9 PCFs ($\deg a = 2$, b a product of four linear factors) with Möbius hashing against standard constants: 239 unidentified FR-candidates, 45 with full FR, 23 outside Conjecture 1.3 and a ${}_3F_2$ scan.

2. Extended search on the six $a(n)$ of these 23: $b(n) = s\prod_{i=1}^{4}(k_i n + t_i)$, $1 \le k_i \le 6$, $|t_i| \le 10$, leading coefficient B with $\alpha^2 + 4B$ a square (α the leading coefficient of a): 5,274 PCFs with full FR.

3. Removed: Conjecture 1.3 including index shifts $-2..2$ (846), slow convergence (15), rational or quadratic limits (313), deflation (3,684); clustering by Möbius equivalence left 347 new clusters; a Cohen-database limit (1), $\pi\sqrt{3}$ (1), ${}_3F_2$ (2).

4. Euler type: an order-one right factor of the recurrence over $\mathbb{Q}(n)$ (Petkovšek's algorithm via ore_algebra): 61 of the extended-search cases and one of the original 23 (Section 3).

5. The operator $L = \theta^2(\theta-1)^2 - z\,a(\theta+1)\theta^2 - z^2\,b(\theta+2)$, $\theta = z\,d/dz$, annihilates $\sum_n Q_n z^n/(n!)^2$; factoring over $\mathbb{Q}(z)$ with ore_algebra, fifteen operators are irreducible, the others factor as $[1,1,1,1]$, $[1,1,2]$ or $[1,3]$.

3 A hidden Euler-type example

Theorem 1. *For* $a(n) = 4n^2+10n-3$ *and* $b(n) = -n^2(n+4)(3n-4)$, $P_n = -(n+1)!\,\prod_{m=0}^{n-1}(3m+2)\,(n^2+3n+3)$ *and*

$$x = \frac{1}{S-1/3}, \qquad S = -\frac{1}{24}\sum_{k\ge 1}\frac{k!\,(k+2)(k+3)(k+4)}{(2/3)_k\,3^k\,(k^2+k+1)(k^2+3k+3)}.$$

Proof sketch. With $r(n) = P_n/P_{n-1} = (n+1)(3n-1)(n^2+3n+3)/(n^2+n+1)$ the identity $a(n)r(n-1)+b(n) = r(n)r(n-1)$ holds in $\mathbb{Q}(n)$, and the initial values agree; the closed form follows by induction. The Casoratian of (P,Q) gives $Q_n/P_n = Q_0/P_0 + \sum_{k=1}^{n}\prod_{j\le k}(-b(j))/(P_kP_{k-1})$, whose limit simplifies to $S-1/3$. Checked to 120 digits. □

The factor $\prod(3m+2)$ explains the gcd pattern (missing powers of 3; the p-adic valuation depends on $3^{-1} \bmod p$), and $n^2+3n+3 = \frac{1}{4}((2n+3)^2+3)$ is a norm form of $\mathbb{Q}(\sqrt{-3})$.

4 Two $\Gamma(1/3)$-type members

With the same $a(n) = 4n^2+10n-3$: for $b(n) = (n+3)(2n-5)(2n-1)(3n-1)$ the limit is a Möbius transform of $\Gamma(1/3)^3\sqrt{3}/\pi$ (match with a limit in Cohen's database); for $b(n) = (n+1)(2n-3)(2n+1)(3n-5)$ one has numerically (250 digits) $x = (27-40c)/(9-24c)$ with $c = \pi^2/\Gamma(1/3)^3$. Both operators factor as $[1,3]$.

5 The fifteen candidates

	$a(n)$	$b(n)$	x	ρ
C0	$4n^2+10n-3$	$(2n-3)\,(2n+1)\,(3n+2)\,(5n-4)$	$-3.9769756756771211392\ldots$	$-39/20$
C1	$4n^2+10n-3$	$(2n+1)\,(2n+5)\,(3n+4)\,(5n-6)$	$-7.6720045669236330443\ldots$	$-71/20$
C2	$3n^2+8n+8$	$2\,(2n+3)\,(3n+1)\,(3n+2)\,(3n+8)$	$33.5147685956496535548\ldots$	$-187/42$
C3	$3n^2+8n+8$	$2\,(2n+3)\,(3n-1)\,(3n+8)\,(5n+3)$	$27.7112473245165813751\ldots$	$-187/45$
C4	$3n^2-2n+2$	$2\,(2n-1)\,(3n-8)\,(3n-2)\,(5n-8)$	$-9.6178657191277317360\ldots$	$3/5$
C5	$4n^2+10n-3$	$(2n-5)\,(2n+3)\,(3n+1)\,(5n+3)$	$192.502805278613431500\ldots$	$-47/20$
C6	$4n^2+9n+1$	$(2n+3)\,(4n+1)\,(4n+9)\,(6n+7)$	$39.4516159585089660963\ldots$	$-123/28$
C7	$3n^2+8n+8$	$(2n+1)\,(3n-2)\,(3n-1)\,(3n+5)$	$9.43457677466345689707\ldots$	$-14/5$
C8	$6n^2+13n+5$	$2\,(2n+1)\,(2n+5)\,(3n+4)\,(3n+7)$	$36.8847177941179069744\ldots$	$-83/18$
C9	$6n^2+13n+5$	$(3n-10)\,(3n-4)\,(3n+10)\,(5n+3)$	$-41.304717113981754226\ldots$	$-121/60$
C10	$3n^2-2n+2$	$(2n-1)\,(3n-2)\,(3n-1)\,(3n+4)$	$2.89364140886627947256\ldots$	$-8/5$
C11	$3n^2-2n+2$	$(3n-5)\,(3n-4)\,(3n+1)\,(4n-3)$	$3.53663600078762561577\ldots$	$-25/84$

C12	$4n^2+10n-3$	$(2n-5)(2n+3)(3n+5)(5n-9)$	$49.6956732085298041815\ldots$	$-39/20$
C13	$6n^2+13n+5$	$2(2n+5)(2n+9)(4n+3)(5n-8)$	$-54.075344587385395944\ldots$	$-241/52$
C14	$3n^2-2n+2$	$(3n-5)(3n+4)(4n-7)(5n-2)$	$27.4046292353155236807\ldots$	$-32/45$

ρ: the non-integer local exponent of L at the finite singularity of smallest modulus.

Tests passed by all fifteen (none identified). Not in Conjecture 1.3 (shifts $-2..2$); not deflatable; limit not rational or quadratic; not Möbius ($|c| \le 10^5$) of 36 standard constants; not Möbius ($|c| \le 6$) of the 635 limits of arXiv:2409.06086v4 or of the evaluable limit expressions of arXiv:2607.06581; not Möbius ($|c| \le 10^4$) of the 171 PCF limits of the euler2ai data set; no hypergeometric solution over $\mathbb{Q}(n)$; operator irreducible over $\mathbb{Q}(z)$; PSLQ $[1, x, c, xc]$ ($|c| \le 10^8$, 1,000 digits) against 21,984 constants (polylogarithms, Dirichlet L-values for $|d| \le 100$, Clausen values, $\Gamma(j/d)$ products with $d \le 24$, complete elliptic integrals at singular moduli); no match in the OEIS for the first 18 decimal digits of $|x|$ or for the first 8 reduced convergent denominators (controls: π and Catalan's constant were found); the two $\Gamma(1/3)$ cases, used as positive controls, were identified by the same code. For all fifteen, x equals the ratio of the two connection coefficients of L at its dominant singularity; the individual coefficients are not Γ-products in an independent basis (150 digits; planted controls recovered).

6 Limitations

Novelty is not established: unpublished databases (e.g. the Ramanujan Library) were not available to us. The LMFDB has no search for decimal constants and was not used. Irreducibility was tested over $\mathbb{Q}(z)$ only. A scheme-guided ${}_4F_3$ test failed its own control and is not counted. Full FR is a numerical criterion.

Data and code

Scripts, logs and the candidate list (SHA-256 time-stamped in github.com/vichienFF/ramanujan-explorer before disclosure) are released with this preprint.

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